



Web-Based Application Development (CSE3023)

Lab Report 8

Group:

K1

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Program:

Bachelor of Computer Science (Software Engineering)
with Honours

Date of Submission: 02 June 2026

Lab Exercise

Using the given database schema, create an MVC Application [CRUD] for Car Shop, using JSP, Servlet, and MySQL.

	#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1	carId	int			No	None		AUTO_INCREMENT	Change Drop More
☐	2	Brand	varchar(15)	utf8mb4_0900_ai_ci		Yes	NULL			Change Drop More
☐	3	Model	varchar(30)	utf8mb4_0900_ai_ci		Yes	NULL			Change Drop More
☐	4	Cyclinder	int			Yes	NULL			Change Drop More
☐	5	Price	double			Yes	NULL			Change Drop More

CarDAO.java

```
7 import java.sql.Connection;
8 import java.sql.DriverManager;
9 import java.sql.PreparedStatement;
10 import java.sql.ResultSet;
11 import java.sql.SQLException;
12 import java.util.ArrayList;
13 import java.util.List;
14 import model.Car;
15
16 public class CarDAO {
17     private final String jdbcURL = "jdbc:mysql://localhost:3306/carshop";
18     private final String jdbcUsername = "root";
19     private final String jdbcPassword = "admin";
20
21     private static final String INSERT_CAR_SQL = "INSERT INTO carpricelist (Brand, Model, Cyclinder, Price) VALUES (?, ?, ?, ?)";
22     private static final String SELECT_CAR_BY_ID = "SELECT carId, Brand, Model, Cyclinder, Price FROM carpricelist WHERE carId = ?";
23     private static final String SELECT_ALL_CAR = "SELECT * FROM carpricelist";
24     private static final String DELETE_CAR_SQL = "DELETE FROM carpricelist WHERE carId = ?";
25     private static final String UPDATE_CAR_SQL = "UPDATE carpricelist SET Brand = ?, Model = ?, Cyclinder = ?, Price = ? WHERE carId = ?";
26
27     public CarDAO() {}
28
29     protected Connection getConnection() {
30         Connection connection = null;
31         try {
32             Class.forName("com.mysql.cj.jdbc.Driver");
33             connection = DriverManager.getConnection(jdbcURL, jdbcUsername, jdbcPassword);
34         } catch (SQLException | ClassNotFoundException e) {
35             e.printStackTrace();
36         }
37         return connection;
38     }
39
40     public void insertCar(Car car) throws SQLException {
41         try (Connection con = getConnection();
42             PreparedStatement ps = con.prepareStatement(INSERT_CAR_SQL)) {
43             ps.setString(1, car.getBrand());
44             ps.setString(2, car.getModel());
45             ps.setInt(3, car.getCyclinder());
46             ps.setDouble(4, car.getPrice());
47             ps.executeUpdate();
48         } catch (SQLException e) {
49             printSQLException(e);
50             throw e;
51         }
52     }
53
54     public Car selectCar(int carId) {
55         Car car = null;
56         try (Connection con = getConnection();
57             PreparedStatement ps = con.prepareStatement(SELECT_CAR_BY_ID)) {
58             ps.setInt(1, carId);
59             try (ResultSet rs = ps.executeQuery()) {
60                 if (rs.next()) {
61                     String brand = rs.getString("Brand");
62                     String model = rs.getString("Model");
63                     int cylinder = rs.getInt("Cyclinder");
64                     double price = rs.getDouble("Price");
65                     car = new Car(carId, brand, model, cylinder, price);
66                 }
67             }
68         } catch (SQLException e) {
```

```

69         printSQLException(e);
70     }
71     return car;
72 }
73
74 public List<Car> selectAllCars() {
75     List<Car> cars = new ArrayList<>();
76     try (Connection con = getConnection();
77         PreparedStatement ps = con.prepareStatement(SELECT_ALL_CAR);
78         ResultSet rs = ps.executeQuery()) {
79
80         while (rs.next()) {
81             int id = rs.getInt("carId");
82             String brand = rs.getString("Brand");
83             String model = rs.getString("Model");
84             int cylinder = rs.getInt("Cylinder");
85             double price = rs.getDouble("Price");
86             cars.add(new Car(id, brand, model, cylinder, price));
87         }
88     } catch (SQLException e) {
89         printSQLException(e);
90     }
91     return cars;
92 }
93
94 public boolean deleteCar(int carId) throws SQLException {
95     boolean rowDeleted;
96     try (Connection con = getConnection();
97         PreparedStatement ps = con.prepareStatement(DELETE_CAR_SQL)) {
98         ps.setInt(1, carId);
99         rowDeleted = ps.executeUpdate() > 0;
100     }
101     return rowDeleted;
102 }
103
104 public boolean updateCar(Car car) throws SQLException {
105     boolean rowUpdated;
106     try (Connection con = getConnection();
107         PreparedStatement ps = con.prepareStatement(UPDATE_CAR_SQL)) {
108         ps.setString(1, car.getBrand());
109         ps.setString(2, car.getModel());
110         ps.setInt(3, car.getCylinder());
111         ps.setDouble(4, car.getPrice());
112         ps.setInt(5, car.getCarId());
113
114         rowUpdated = ps.executeUpdate() > 0;
115     }
116     return rowUpdated;
117 }
118
119 private void printSQLException(SQLException ex) {
120     for (Throwable e : ex) {
121         if (e instanceof SQLException) {
122             e.printStackTrace(System.err);
123             System.err.println("SQLState: " + ((SQLException) e).getSQLState());
124             System.err.println("Error Code: " + ((SQLException) e).getErrorCode());
125             System.err.println("Message: " + e.getMessage());
126
127             Throwable t = ex.getCause();
128             while (t != null) {
129                 System.out.println("Cause: " + t);
130                 t = t.getCause();
131             }
132         }
133     }
134 }
135 }

```

CarServlet.java

```
5  package web;
6
7  import java.io.IOException;
8  import javax.servlet.ServletException;
9  import javax.servlet.annotation.WebServlet;
10 import javax.servlet.http.HttpServlet;
11 import javax.servlet.http.HttpServletRequest;
12 import javax.servlet.http.HttpServletResponse;
13
14 import model.Car;
15 import dao.CarDAO;
16 import java.sql.SQLException;
17 import java.util.List;
18 import javax.servlet.RequestDispatcher;
19
20 @WebServlet("/")
21 public class CarServlet extends HttpServlet {
22
23     private CarDAO carDAO;
24
25     @Override
26     public void init() {
27         carDAO = new CarDAO();
28     }
29
30     @Override
31     protected void doGet(HttpServletRequest request, HttpServletResponse response)
32         throws ServletException, IOException {
33         String action = request.getServletPath();
34
35         try {
36             switch (action) {
37                 case "/list":
38                 case "/":
39                     listCar(request, response);
40                     break;
41                 case "/new":
42                     showNewForm(request, response);
43                     break;
44                 case "/insert":
45                     insertCar(request, response);
46                     break;
47                 case "/delete":
48                     deleteCar(request, response);
49                     break;
50                 case "/edit":
51                     showEditForm(request, response);
52                     break;
53                 case "/update":
54                     updateCar(request, response);
55                     break;
56                 default:
57                     listCar(request, response);
58                     break;
59             }
60         } catch (SQLException ex) {
61             throw new ServletException(ex);
62         }
63     }
64
65     private void listCar(HttpServletRequest request, HttpServletResponse response)
66         throws SQLException, IOException, ServletException {
67         List<Car> listCar = carDAO.selectAllCars();
68         request.setAttribute("list", listCar);
```

```

69     RequestDispatcher dispatcher = request.getRequestDispatcher("CarList.jsp");
70     dispatcher.forward(request, response);
71 }
72
73 private void showNewForm(HttpServletRequest request, HttpServletResponse response)
74     throws ServletException, IOException {
75     RequestDispatcher dispatcher = request.getRequestDispatcher("CarForm.jsp");
76     dispatcher.forward(request, response);
77 }
78
79 private void showEditForm(HttpServletRequest request, HttpServletResponse response)
80     throws ServletException, IOException {
81     int carId = Integer.parseInt(request.getParameter("carId"));
82     Car existingCar = carDAO.selectCar(carId);
83
84     request.setAttribute("car", existingCar);
85     RequestDispatcher dispatcher = request.getRequestDispatcher("CarForm.jsp");
86     dispatcher.forward(request, response);
87 }
88
89 private void insertCar(HttpServletRequest request, HttpServletResponse response)
90     throws ServletException, IOException {
91     String brand = request.getParameter("Brand");
92     String model = request.getParameter("Model");
93
94     int cylinder = Integer.parseInt(request.getParameter("Cylinder"));
95     double price = Double.parseDouble(request.getParameter("Price"));
96
97     Car newCar = new Car(brand, model, cylinder, price);
98     carDAO.insertCar(newCar);
99
100     response.sendRedirect(request.getContextPath() + "/");
101 }
102
103 private void updateCar(HttpServletRequest request, HttpServletResponse response)
104     throws ServletException, IOException {
105     int carId = Integer.parseInt(request.getParameter("carId"));
106     String brand = request.getParameter("Brand");
107     String model = request.getParameter("Model");
108     int cylinder = Integer.parseInt(request.getParameter("Cylinder"));
109     double price = Double.parseDouble(request.getParameter("Price"));
110
111     Car car = new Car(carId, brand, model, cylinder, price);
112     carDAO.updateCar(car);
113
114     response.sendRedirect(request.getContextPath() + "/");
115 }
116
117 private void deleteCar(HttpServletRequest request, HttpServletResponse response)
118     throws ServletException, IOException {
119     int carId = Integer.parseInt(request.getParameter("carId"));
120     carDAO.deleteCar(carId);
121
122     response.sendRedirect(request.getContextPath() + "/");
123 }
124
125 @Override
126 protected void doPost(HttpServletRequest request, HttpServletResponse response)
127     throws ServletException, IOException {
128     doGet(request, response);
129 }
130 }

```

Car.java

```
5  package model;
6
7  /**
8   *
9   * @author Asus
10  */
11  public class Car {
12
13      private int carId;
14      private String Brand;
15      private String Model;
16      private int Cyclinder;
17      private double Price;
18
19      public Car() {
20
21      }
22
23      public Car (int carId, String b, String m, int c, double p){
24          this.carId = carId;
25          this.Brand = b;
26          this.Cyclinder = c;
27          this.Model = m;
28          this.Price = p;
29      }
30
31      public Car (String b, String m, int c, double p){
32          this.Brand = b;
33          this.Cyclinder = c;
34          this.Model = m;
35
36          this.Price = p;
37      }
38
39      public int getCarId() {
40          return carId;
41      }
42
43      public void setCarId(int carId) {
44          this.carId = carId;
45      }
46
47      public String getBrand() {
48          return Brand;
49      }
50
51      public void setBrand(String Brand) {
52          this.Brand = Brand;
53      }
54
55      public String getModel() {
56          return Model;
57      }
58
59      public void setModel(String Model) {
60          this.Model = Model;
61      }
62
63      public int getCyclinder() {
64          return Cyclinder;
65      }
66
67      public void setCyclinder(int Cyclinder) {
68          this.Cyclinder = Cyclinder;
69      }
```

```

69
70     public double getPrice() {
71         return Price;
72     }
73
74     public void setPrice(double Price) {
75         this.Price = Price;
76     }
77
78 }
79

```

CarIndex.jsp

```

1  <%--
2      Document    : CarIndex
3      Created on  : 2 Jun 2026, 5:11:01 pm
4      Author     : Asus
5  --%>
6  <%@page contentType="text/html" pageEncoding="UTF-8"%>
7  <!DOCTYPE html>
8  <html>
9      <head>
10         <meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
11         <title>Car Management Application</title>
12         <link rel="stylesheet" href="https://stackpath.bootstrapcdn.com/bootstrap/4.3.1/css/bootstrap.min.css">
13     </head>
14     <body class="container mt-5">
15         <div class="jumbotron">
16             <h1>Application MVC system for Car Management</h1>
17             <br>
18             <div class="list-group col-md-4">
19                 <a class="list-group-item list-group-item-action" href="${pageContext.request.contextPath}/list" >All Car List</a>
20                 <a class="list-group-item list-group-item-action" href="${pageContext.request.contextPath}/new" >Add a New Car</a>
21             </div>
22         </div>
23     </body>
24 </html>

```

Carform.jsp

```

6  <%@page language="java" contentType="text/html" pageEncoding="UTF-8"%>
7  <%@taglib uri="http://java.sun.com/jsp/jstl/core" prefix="c" %>
8  <!DOCTYPE html>
9  <html>
10     <head>
11         <meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
12         <title>Car Management Application</title>
13         <link rel="stylesheet" href="https://stackpath.bootstrapcdn.com/bootstrap/4.3.1/css/bootstrap.min.css">
14     </head>
15     <body>
16         <header>
17             <nav class="navbar navbar-expand-md navbar-dark" style="background-color: tomato">
18                 <div>
19                     <a href="${pageContext.request.contextPath}/" class="navbar-brand">Car Management App</a>
20                 </div>
21                 <ul class="navbar-nav">
22                     <li><a href="${pageContext.request.contextPath}/" class="nav-link">Car List</a></li>
23                 </ul>
24             </nav>
25         </header>
26         <br>
27         <div class="container col-md-5">
28             <div class="card">
29                 <div class="card-body">
30                     <form action="${pageContext.request.contextPath}/${car != null ? 'update' : 'insert'}" method="post">
31
32                         <h2>
33                             <c:out value="${car != null ? 'Edit Car' : 'Add New Car'}" />
34                         </h2>

```

```

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47
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49
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53
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58
59
60
61
62
63
64
65
66
<c:if test="${car != null}">
  <input type="hidden" name="carId" value="${c:out value='${car.carId}' }/" />
</c:if>

<fieldset class="form-group">
  <label>Car Brand</label>
  <input type="text" value="${c:out value='${car.brand}' }/" class="form-control" name="Brand" required="required">
</fieldset>

<fieldset class="form-group">
  <label>Car Model</label>
  <input type="text" value="${c:out value='${car.model}' }/" class="form-control" name="Model" required="required">
</fieldset>

<fieldset class="form-group">
  <label>Car Cylinder</label>
  <input type="number" value="${c:out value='${car.cylinder}' }/" class="form-control" name="Cylinder" required="required">
</fieldset>

<fieldset class="form-group">
  <label>Car Price</label>
  <input type="text" value="${c:out value='${car.price}' }/" class="form-control" name="Price" required="required">
</fieldset>

<button type="submit" class="btn btn-success">Save</button>
</form>
</div>
</div>
</div>
</body>
</html>

```

CarList.jsp

```

1
2
3
4
5
6
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9
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12
13
14
15
16
17
18
19
20
21
22
23
24
25
26
27
28
29
30
31
32
33
34
<%@page language="java" contentType="text/html" pageEncoding="UTF-8"%>
<%@ taglib uri="http://java.sun.com/jsp/jstl/core" prefix="c"%>
<!DOCTYPE html>
<html>
  <head>
    <meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
    <title>Car Management Application</title>
    <link rel="stylesheet" href="https://stackpath.bootstrapcdn.com/bootstrap/4.3.1/css/bootstrap.min.css">
  </head>
  <body>
    <header>
      <nav class="navbar navbar-expand-md navbar-dark" style="background-color: tomato">
        <div>
          <a href="${pageContext.request.contextPath}/" class="navbar-brand">Car Management App</a>
        </div>
        <ul class="navbar-nav">
          <li><a href="${pageContext.request.contextPath}/" class="nav-link">Car List</a></li>
        </ul>
      </nav>
    </header>
    <br>
    <div class="row">
      <div class="container">
        <h3 class="text-center">List of Cars</h3>
        <hr>
        <div class="container text-left">
          <a href="${pageContext.request.contextPath}/new" class="btn btn-success">Add New Car</a>
        </div>
        <br>
        <table class="table table-bordered">
          <thead>
            <tr>
              <th>ID</th>
              <th>Brand</th>

```

[illegible]

Output:

Application MVC system for Car Management

All Car List

[Add a New Car](#)

Add New Car

Car Brand

Car Model

Car Cylinder

Car Price

[Save](#)

List of Cars

Add New Car

ID	Brand	Model	Cylinder	Price	Actions
1	a	a	4	\$1500.0	Edit Delete
2	a	a	6	\$1500.0	Edit Delete
3	toyota	v3	2	\$1300000.0	Edit Delete

Update

Edit Car

Car Brand

toyota

Car Model

v2

Car Cylinder

2

Car Price

1300000.0

Save

List of Cars

Add New Car

ID	Brand	Model	Cylinder	Price	Actions
1	a	a	4	\$1500.0	Edit Delete
2	a	a	6	\$1500.0	Edit Delete
3	toyota	v2	2	\$1300000.0	Edit Delete

Delete

[YouTube](#) [Translate](#) [Instagram](#) [Telegram](#)

Car List

localhost:8080 says

Are you sure you want to delete this car?

OK

Cancel

Add New Car

ID	Brand	Model	Cylinder	Price	Actions
1	a	a	4	\$1500.0	Edit Delete
2	a	a	6	\$1500.0	Edit Delete
3	toyota	v2	2	\$1300000.0	Edit Delete

List of Cars

Add New Car

ID	Brand	Model	Cylinder	Price	Actions
1	a	a	4	\$1500.0	Edit Delete
2	a	a	6	\$1500.0	Edit Delete